6.3 Adding Integers Using Integer Chips

Lesson Introduction

Benchmark	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.	Learning Targets	• I can express integer addition using integer chips. • I can add integers using integer chips.

	Building On	Working Toward
Benchmark Coherence	MA.5.NSO.2 Add, subtract, multiply, and divide multidigit numbers.	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

Essential Questions	• What is an integer? • How can you use integer chips to represent integer addition? • What is a zero pair? • How do you use integer chips to add integers?
Lesson Narrative	In this lesson, students represent integer addition using integer chips to begin developing an understanding of how to add signed numbers. Students explore zero pairs and begin generalizing about patterns they observe when adding integers with the same sign or opposite signs.

Component Summary	6.3.1 Warm-Up (~5 min.)	Make connections between elevations and positive and negative integers *(SE p. 243)*	6.3.4 Bring It Together (5 min.)	Summarize how to add integers with the same sign using integer chips *(SE p. 245)*
	6.3.2 Guided Instruction (5-10 min.)	Represent simple money contexts with positive and negative integers and integer chips and define zero pairs *(SE p. 243)*	6.3.5 Exploration (15 min.)	Explore using zero pairs to add integers with opposite signs using integer chips *(SE p. 246)*
	6.3.3 Exploration (10 min.)	Explore adding integers with the same sign using integer chips *(SE p. 245)*	6.3.6 Bring It Together (5 min.)	Summarize how to add integers with opposite signs *(SE p. 248)*
	6.3.7 Wrap-Up (~5 min.)	Demonstrate understanding of the lesson learning targets *(SE p. 248)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Integers • Zero pair • Zero pair as a mathematical equation Additional Terms: N/A	• Integer chips • (Optional) $1 manipulatives	• Consider creating sentence stems and/or answer choices for question 2 in 6.3.6 Bring It Together.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may struggle to come up with real-world scenarios to represent addition expressions, particularly when addends are negative.</td><td>Consider providing integer chips for students in 6.3.5 Exploration or encouraging students to draw a representation if they get stuck.
Consider using $1 manipulatives in 6.3.3 Exploration to help students connect the idea of spending and earning money to positive and negative integers.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
In 6.3.3 Exploration and 6.3.5 Exploration, provide an opportunity for students to notice and wonder about patterns in the sums when adding integers with the same or opposite signs.</td><td>MA.K12.MTR.2.1: Multiple Representations
In 6.3.3 Exploration and 6.3.5 Exploration, students have opportunities to represent adding integers using multiple representations. Students make connections among written descriptions of real-world scenarios, numeric expressions, and integer chips.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Lesson 3 incorporates integer chips into instruction to provide meaningful learning strategies for making sense of adding integers. Provide integer chips for students to explore each activity throughout the lesson.
<table><tr><th>Earn</th><th>Spend</th></tr><tr><td>+</td><td>−</td></tr></table></td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.3.1 Warm-Up: Below Sea Level

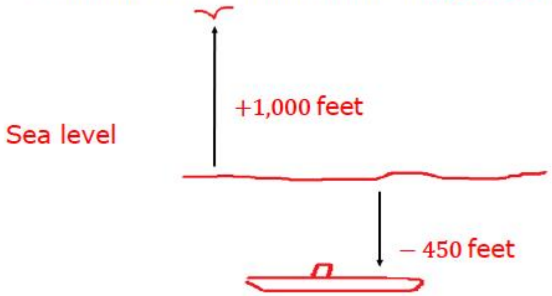
Component Overview: *Students connect positive and negative integers to distances above and below sea level and sketch an example.*

6.3.1 Warm-Up: Below Sea Level

Dr. Ernest Everett Just was a pioneering African American biologist who conducted research at Woods Hole Marine Biological Laboratory in Massachusetts. He published two books and more than 70 scientific papers, most of which focused on the cell. Today, marine biologists use submarines to study marine life.

A submarine explorer begins at sea level and descends at a rate of 150 feet per minute.

1. Sketch a picture that shows how far below sea level the submarine will be in 3 minutes. *Answers will vary.*



2. Add a bird that is flying 1,000 feet above sea level to your picture. *See above*
3. What number represents sea level?
   0
4. The location of the submarine is represented by a [○ positive value, / ○ negative value,] while the location of the bird is represented by a [○ positive value. / ○ negative value.]

Dr. Ernest Everett Just accomplished many things in his career that may pique students' interests in the STEM field. Consider an interdisciplinary extension project with a science teacher on one of his many roles.

As students work on question 1, consider asking them the following:

- *Does your sketch more closely resemble a horizontal or vertical number line? Why?*
- *What value does sea level represent in your sketch?*

6.3.2 Guided Instruction: Zero Pairs

Component Overview: *Students learn how positive and negative integers can be represented using integers and integer chips. The concept of zero pairs is introduced in the context of adding integers. Students review the definition and then use zero pairs to add integers.*

6.3.2 Guided Instruction: Zero Pairs

Positive and negative values are often useful in the contexts of money, banking, and accounting.

- In the context of money, ○ positive ○ negative numbers can be used to represent money spent or money owed to someone.
- ○ Positive ○ Negative numbers can be used to represent money earned.

Money can be used to learn how to add integers.

- A value of $+1$ means that you *earned* $\$1$.
- A value of (-1) means that you *spent* $\$1$ or *owe* $\$1$.

The table below shows a positive integer chip, which represents earning $\$1$, and a negative integer chip, which represents spending $\$1$.

Earn	Spend
+	−

Integer chips can also be used to represent integers.

Integers are whole numbers and their opposites.

Consider providing a few real-world examples of positive and negative numbers representing money. These examples will activate prior knowledge and help students connect spending and earning to positive and negative numbers.

6.3.2 Guided Instruction: Zero Pairs (continued)

A positive number can be written with a positive sign, $+1$ or $(+1)$, or it can be written without any sign, 1.

A negative number can be written as -1 , (-1) or $^{-}1$.

1. Complete the table. In the last column, sketch the integer chips that represent the phrase and the integer. Place a + sign in a circle to represent ⊕ and place a — sign in a circle to represent ⊖.

Phrase	Integer	Integer Chips
Earned $6	$(+6)$	⊕⊕⊕⊕⊕⊕
Spent $3	-3	⊖⊖⊖

2. Complete the table by sketching integer chips for each phrase.

Earned $4	Spent $4	Earned $4, Then Spent $4
⊕⊕⊕⊕	⊖⊖⊖⊖	⊕⊕⊕⊕ ⊖⊖⊖⊖

3. Joleen earned $4, then spent $4. How much money does she have left?
   0

The last column in the table and problem above both represent a ***zero pair***.

A ***zero pair*** is a pair of numbers whose sum is zero.

Consider the tools available for instruction during this activity. Modeling each example using inquiry would benefit students as they make connections from the integer-chip representations to their numerical representations.

Consider asking students the following:

- *Can you explain what a zero pair is in your own words?*
- *Can you provide an example of a zero pair?*

4. Complete the table by drawing the negative integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
+ + +	− − −	+ + + − − −

Each zero pair can be represented by an equation. The box shows two mathematical equations that represent the zero pair of + and −.

Zero Pair as a Mathematical Equation

$^{+}1 + (-1) = 0$ or $1 + {}^{-}1 = 0$

5. Write an equation that represents the **sum** column in the previous table.
$+3 + (-3) = 0$

6. Complete the table by drawing the positive integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
+ +	− −	+ + − −

7. Write an equation that represents the **sum** column in the previous table.
$+2 + (-2) = 0$

Students are only given the visual model for positive integer chips. They complete the visual model for negative integer chips and the sum.

To challenge advanced learners, consider having students write a word problem to go with the equations they have created to represent zero pairs.

Students are only given the visual model for negative integer chips. They complete the visual model for positive integer chips and the sum.

6.3.3 Exploration: Adding Integers with the Same Sign

Component Overview: *Students work with a partner to complete a table that includes multiple representations of adding integers with the same sign. The table includes the numeric expression, money sentence, integer chips, and sum. Next, students draw conclusions about patterns they notice about the signs of the sums and if there are any zero pairs when adding integers with the same sign.*

6.3.3 Exploration: Adding Integers with the Same Sign

1. Work with your partner to complete the table.

Expression	Money Sentence	Integer Chips	Sum
$5+2$	I earn \$5 babysitting and another \$2 for cleaning the dishes.	⊕⊕⊕⊕⊕ ⊕⊕	7
$3+2$	*Answers will vary. I earn \$3 mowing the lawn and another \$2 for taking out the trash. I earn \$3 washing dishes and another \$2 for babysitting.*	⊕⊕⊕ ⊕⊕	5
$^{-}4 + {}^{-}3$	*Answers will vary. I owe my mom \$4 and I spent \$3 at the dollar store. I loaned my friend \$4 and I spent \$3 on my mom's birthday present.*	⊖⊖⊖⊖ ⊖⊖⊖	-7
$(-2)+(-1)$	*Answers will vary. I spent \$2 at the candy store and I paid back my mom the \$1 I owed her. I bought a \$2 ice cream cone and I loaned my sister \$1.*	⊖⊖ ⊖	-3

2. With your partner, discuss the patterns you notice about the signs of the sums when adding integers with the same sign.
   Answers will vary. I notice that when the signs were the same, we added the total number of chips.
3. Were any zero pairs created when adding integers with the same sign? Explain.
   Answers will vary. No zero pairs were created when adding integers with the same sign. Instead, the result was a larger quantity of the signed integers.

Students are only given the visual model of integer chips for the first row.

Notice and Wonder
Consider engaging students with a discussion regarding what they notice and wonder about the signs and if there were any zero pairs prior to them writing their responses to questions 2 and 3.

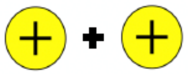
Consider asking students the following:
- *What is the sign of the sum of two positive integers?*
- *What is the sign of the sum of two negative integers?*

6.3.4 Bring It Together: Adding Integers with the Same Sign

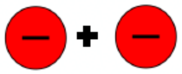
Component Overview: *Students summarize how to add integers with the same sign using integer chips. Consider creating an anchor chart and displaying it for students to reference.*

6.3.4 Bring It Together: Adding Integers with the Same Sign

1. Complete the visual model of the rule of adding two positive integers by drawing the integer chip that represents the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

2. Complete the visual model of the rule of adding two negative integers by drawing the integer chip that represent the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

Prompt students to justify their visual models to a partner. Highlight any students that make changes based on their partner's feedback to encourage growth mindset.

Consider asking students the following:

- *Will the sum of two positive integers always be positive?*
- *Will the sum of two negative integers always be negative?*

To add complexity, consider asking students who finish early to write equations to match each visual model of addition.

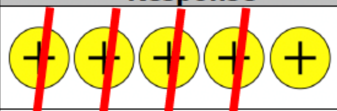
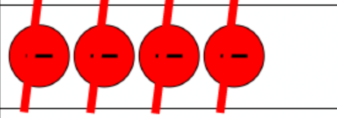
6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

Component Overview: *Students use integer chips to represent the sum of integers with opposite signs, determining the sum based on the zero pairs. Students then work with a partner to continue adding integers with opposite signs and create money sentences to represent addition expressions. Students work independently to find the sum of integers, check their partner's work, and use the sums to generalize how to add integers with opposite signs.*

6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

It is possible to add a positive number and a negative number. To find the answer using integer chips, combine a + and − to create a zero pair. Make as many zero pairs as possible. Then, determine the number of chips that are left over.

1. Complete the table to show how the sum $5 + (^-4)$ can be represented using integer chips.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	*4 zero pairs*
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the **sum** using the number of integer chips remaining.	$5 + (^-4) = 1$

2. What do you notice about the remaining integer chips and the sign of the sum?
There was 1 positive integer chip remaining. The sum was 1.

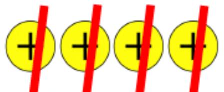
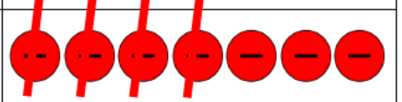
Consider asking students the following:

- *How can you represent 5 with integer chips?*
- *How can you represent -4 using integer chips?*
- *Why is the sum of 5 and -4 positive?*

For both questions 1 and 3, students are given the visual model for the first two steps and must complete the third row in addition to synthesis questions 2 and 4.

6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs (continued)

3. Complete the table independently to show how the sum $4 + (-7)$ can be represented using integer chips. Check your work with your partner.

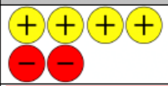
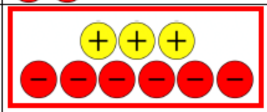
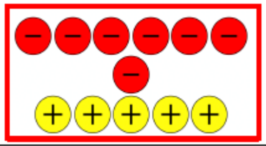
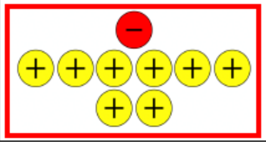
Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	*4 zero pairs*
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the **sum** using the number of integer chips remaining.	$4 + (-7) = -3$

4. What do you notice about the remaining integer chips and the sign of the sum?
There were 3 negative integer chips remaining. The sum was -3.

Consider having students discuss what they think the sign of the sums will be based on what they notice about the addition expressions before having them complete the table for question 5. Guide students to make logical predictions based on what they know about adding signed numbers and absolute value.

6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs (continued)

5. Work with your partner to complete the table. The first row is completed.

Expression	Money Sentence	Integer Chips	Sign of Sum	Sum
$4+(-2)$	I earn \$4 and spend \$2.		$+$	$+2$
$3+(-6)$	*I earn \$3 and spend \$6.*		$-$	-3
$^{-}7+5$	*I spend \$7 and earn \$5.*		$-$	-2
$(-1)+8$	*I spend \$1 and earn \$8.*		$+$	$+7$

6. Complete the table below independently. Check your work with your partner.

Mathematical Expression	Number of Zero Pairs	Sign of Sum	Sum
$(-10)+12$	10	$+$	2
$12+-7$	7	$+$	5
$(-11)+4$	4	$-$	-7
$8+{}^{-}8$	8	No sign	0
$^{-}10+3$	3	$-$	-7
$9+(-2)$	2	$+$	7

7. What do you notice about expressions where the sum is a positive number?
   Answers will vary. The number with the larger absolute value in the mathematical expression is positive.

8. What do you notice about expressions where the sum is a negative number?
   Answers will vary. The number with the larger absolute value in the mathematical expression is negative.

Consider having students who finish early represent each expression in an additional way, either with an equivalent expression, a visual representation, or a written scenario.

Turn and Talk
Ask students to verbalize their responses before completing their written explanation to questions 7 and 8. This will solidify student understanding and allow for any misconceptions to be resolved before the 6.3.7 Wrap-Up.

What do you notice about the sum when the addend with the larger absolute value is negative?

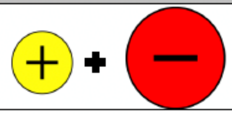
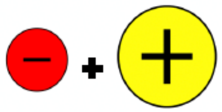
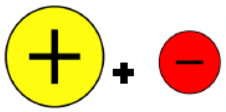
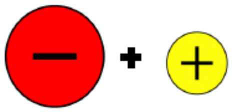
6.3.6 Bring It Together: Adding Integers with the Opposite Sign

Component Overview: *Students write expressions to represent a visual model of addition of integers with opposite signs. Then, students explain two things they noticed about adding integers with opposite signs.*

6.3.6 Bring It Together: Adding Integers with Opposite Signs

The table below shows visual models of rules for adding integers with opposite signs. In each visual model, the larger circle symbolizes the integer represented by the most integer chips.

1. Write an example of the rule. The first row has been completed.

Visual Model of Addition	Sign of Sum	Example
		$2 + (-5) = -3$
		*Answers will vary.* $-3 + 10 = 7$ $-5 + 7 = 2$
		*Answers will vary.* $6 + (-4) = 2$ $10 + (-6) = 4$
		*Answers will vary.* $-9 + 8 = -1$ $-12 + 4 = 8$

2. Give two observations about the rules for adding integers with opposite signs.
   Answers will vary.
   When adding integers with opposite signs:
   - *Ignoring the signs of the numbers in the mathematical expression, subtract the numbers.*
   - *The sign of the sum will be the sign of the number with the greater absolute value in the mathematical expression.*

Consider asking students the following:
- *What do you notice about the size of these integer chips?*
- *How does the absolute value of each addend impact the sign of the sum?*

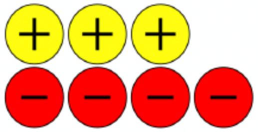
Common Misconception
Monitor for students who may articulate their answers with phrases like "the bigger number will always be larger" or "the answer will take the sign of the bigger number." The sum will take the sign of the addend with the larger absolute value, which is not the larger number if it is negative.

6.3.7 Wrap-Up: Ticket Out the Door

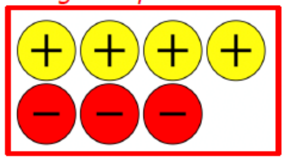
Component Overview: *Students demonstrate their understanding of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.*

6.3.7 Wrap-Up: Ticket Out the Door

1. Simone created the sketch below to represent the expression $4 + (-3)$.



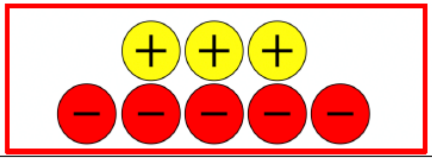
Her sketch is incorrect. Explain why and create a new sketch to find the sum.
There should be 4 positive integer chips and 3 negative integer chips.



Consider providing integer chips to provide a kinesthetic entry point.

Also, consider providing sentence starters for question 2 to help students who struggle writing explanations.

2. Complete the table for the expression below.

Expression	$3 + {}^{-}5$
Integer Chips	
Money Sentence	*Answers will vary. I earned \$3 mowing the lawn and spend \$5 on pizza.* *I earned \$3 taking out the garbage and spend \$5 on candy.*
Sign of the Sum	$-$
Sum	-2